Department of Physics

Semester IV Major Project Report

Dynamics of Self-Propelled Particles in Simple Fluid Flows

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"I'm incredibly grateful to my guide, Dr. Akshay Bhatnagar, for his genuine support and encouragement throughout this project. His knowledge and insights helped me navigate through challenges, and his guidance made all the difference. I would also like to extend my heartfelt thanks to my family and friends for their constant support and encouragement throughout my studies."

Abstract

This project studies the motion of self-propelled particles in a two-dimensional flow field. The particles are not perfect spheres, but are ellipsoidal in shape, and their orientation changes both due to the fluid forces and due to a constant turning tendency called chirality. Random changes in direction, caused by collisions with water molecules, are also included as rotational noise. The goal is to understand how the shape of the particle, chirality, and the properties of the flow affect the way particles move and spread in space. Different combinations of particle parameters and flow conditions were tested using numerical simulations. Results show that shape and chirality play an important role in how the particles explore the flow. When the flow changes with time, the particles are able to escape vortex regions more easily and cover more of the space. The study also looks at the distribution of particles using vorticity.

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1 Introduction

Active matter refers to systems made up of self-driven units that use energy to move and interact with their surroundings. These systems are found in both nature and artificial setups. Examples include swimming bacteria, sperm cells, algae, and synthetic microswimmers. What makes active matter interesting is that the particles do not just passively follow the forces acting on them, but can produce motion on their own. This often leads to complex and interesting patterns of motion, both for individual particles and for groups.

A key part of understanding active systems is looking at how the particles interact with external fields, especially fluid flows. Flows can guide or trap, depending on the flow structure and the properties of the particles.

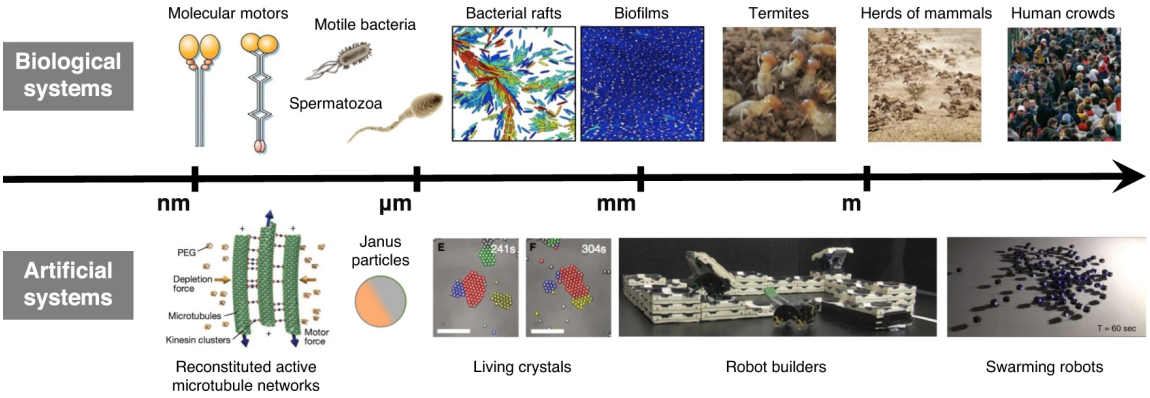


Figure 1: Examples of active matter systems across different length scales, ranging from nanometer-sized molecular motors to human crowds. Both biological and artificial active matter systems show collective behaviors driven by self-propulsion and interactions with their environment. Taken from [1].

In the mini project , the motion of spherical self-propelled particles was studied inside a two-dimensional laminar flow field. The goal was to understand how self-propulsion and the flow work together to control the particle's motion. The spreading of particles was measured using Mean Squared Displacement (MSD). At short times, the motion looked like free-particle motion, but at longer times, the flow traps particles inside vortex structures, and the MSD stops growing. It was also found that increasing the self-propulsion speed makes the motion more chaotic and sensitive to the starting conditions, which was confirmed by calculating Lyapunov exponents.

That work only considered spherical particles. But in real systems, most swimmers are not perfect spheres. Many of them have elongated shapes, which change how they align and move in a flow. In this project, the model is extended to ellipsoidal self-propelled particles, to understand how the shape affects the motion.

Another important factor is chirality. Chirality is modeled as a constant turning tendency, which breaks the symmetry of the motion and changes how particles explore the flow.

Finally, the model also includes rotational diffusion, which represents random changes in particle orientation caused by collisions with water molecules. This effect is especially important for microscopic particles, as it can help them escape traps or prevent long-term alignment with the flow.

The aim of this work is to study the effect of particle shape, chirality, and noise on the motion of self-propelled particles in a two-dimensional flow field.

2 Mathematical Model

2.1 Governing Equations

The dynamics of a self-propelled ellipsoidal particle in a fluid flow are given by:

$$\dot{\vec{r}} = \vec{v}_f(\vec{r}, t) + v_p \hat{p} \tag{1}$$

$$\dot{\hat{p}} = \frac{1}{2}(\vec{\omega} \times \hat{p}) + \alpha(\hat{p} \cdot \mathbf{E})(\mathbf{I} - \hat{p}\hat{p}) \tag{2}$$

The motion of the particle is described using these two equations. The first equation describes how the position changes, and the second describes how the particle's orientation changes over time.

- $\vec{r}$: The position of the particle .
- $\hat{p}$: The unit vector representing the direction the particle is facing or moving.
- $\vec{v}_f(\vec{r}, t)$: The velocity of the fluid at the particle's position at time t.
- v_p: The self-propulsion speed of the particle.
- $\vec{\omega} = \nabla \times \vec{v}_f$: The vorticity of the fluid flow.
- $\mathbf{E} = \frac{1}{2}(\nabla \vec{v}_f + (\nabla \vec{v}_f)^T)$: The strain rate tensor.
- α: A parameter that represents the particle's eccentricity. It is defined as

$$\alpha = \frac{\gamma^2 - 1}{\gamma^2 + 1}$$

  where γ is the ratio of the major axis to the minor axis of the spheroidal particle. For a perfect sphere, $\gamma = 1$ which makes $\alpha = 0$. As the particle becomes more elongated, γ increases, and so does α.

2.2 Representation in 2D

In 2D, we describe orientation with angle θ, and the equations reduce to:

$$\dot{x} = v_x(x, y, t) + v_p \cos\theta \tag{3}$$

$$\dot{y} = v_y(x, y, t) + v_p \sin\theta \tag{4}$$

$$\dot{\theta} = -\alpha \cos x \cos y \sin(2\theta) + \sin x \sin y \tag{5}$$

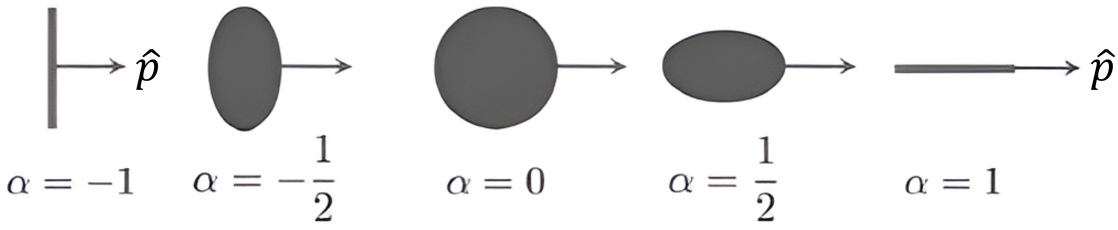


Figure 2: The parameter α characterizes the shape of the ellipse in relation to its swimming direction.[4]

For a general spheroid, the alignment parameter α is given by:

$$\alpha = \frac{\gamma^2 - 1}{\gamma^2 + 1},$$

where γ is the ratio of the major and minor axes of the spheroid. This parameter α accounts for the anisotropy in particle shape, influencing the alignment behavior within the flow.

2.3 Flow Field

The cellular flow is generated via the streamfunction:

$$\psi(x, y, t) = \frac{U_0 L}{2\pi} \sin\left(\frac{2\pi(x - B\sin\nu t)}{L}\right) \sin\left(\frac{2\pi y}{L}\right) \tag{6}$$

The velocity components u_x and u_y in the x- and y-directions, respectively, are derived from the stream function $\psi(x, y)$ using the standard relations:

$$v_x = \partial_y \psi, \quad v_y = -\partial_x \psi \tag{7}$$

The velocity field components for this flow are:

$$v_x(x, y, t) = U_0 \sin\left(\frac{2\pi(x - B\sin\nu t)}{L}\right) \cos\left(\frac{2\pi y}{L}\right) \tag{8}$$

$$v_y(x, y, t) = -U_0 \cos\left(\frac{2\pi(x - B\sin\nu t)}{L}\right) \sin\left(\frac{2\pi y}{L}\right) \tag{9}$$

- B: Amplitude of the oscillation in the x-direction of the stream function.
- U_0: The characteristic velocity of the flow. This sets the overall strength of the fluid's motion in the system.
- L: The characteristic length scale of the system, which also defines the size of the repeating flow pattern.
- ν: Frequency of the time-dependent oscillation.

2.4 Effect of Shape on Aggregation

For spherical particles (where the shape parameter $\alpha = 0$), the equations for motion follow the flow of the fluid closely. The system also satisfies an important condition: the total phase space volume is conserved.

In this case, the full phase space divergence is:

$$\nabla \cdot \mathbf{V} = \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_\theta}{\partial \theta}$$

For $\alpha = 0$, the velocity field (time-independent) is:

$$v_x = U_0 \sin(x)\cos(y), \quad v_y = -U_0 \sin(y)\cos(x), \quad v_\theta = U_0 \sin(x)\sin(y)$$

Taking the divergence:

$$\nabla \cdot \mathbf{V} = U_0 \cos(x)\cos(y) - U_0 \cos(x)\cos(y) + 0 = 0$$

So the phase space volume remains constant. Because of this, particles do not form clusters, and the distribution stays uniform as long as the initial setup is uniform.

When the shape parameter is non-zero ($\alpha \neq 0$), the particle orientation interacts with the flow through an alignment term. In this case, the equation for θ has an extra part:

$$v_\theta = U_0 \sin(x)\sin(y) - \alpha \cos(x)\cos(y)\sin(2\theta)$$

The phase space divergence becomes:

$$\nabla \cdot \mathbf{V} = -\alpha \cos(x)\cos(y)\cos(2\theta)$$

So the divergence is no longer zero. This means that the flow in phase space can contract or expand. Because of this, particles can start to gather (aggregate) in certain areas of the flow.

Simulations show that for non-spherical particles, the density is no longer uniform. Particles tend to avoid the centers of the vortex cells and slowly collect near the separatrices — the boundary regions that separate the vortices.

3 Numerical Methodology

The particle dynamics are evolved using a second-order Runge-Kutta (RK2) integration scheme.The entire code including the integrator and plotting , are documented in a Jupyter Notebook. This notebook is attached along with the report for reference.

All simulations are carried out with a fixed timestep of $dt = 0.01$, and the total integration time is typically set to $t = 200$, unless otherwise specified. The flow field is defined over a periodic domain of size $L = 2\pi$, with the characteristic flow velocity amplitude $U_0 = 1$. Rotational noise is included at each timestep by adding a random angular increment drawn from a normal distribution, scaled by $\sqrt{2D_r\,dt}$, where D_r is the rotational diffusion constant.

The system is explored with different combinations of the following parameters:

- α (particle eccentricity)

- v_p (self-propulsion speed)
- C (chirality strength)
- B, ν (oscillating flow parameters)

4 Results

The numerical simulations were performed using the model described in the previous sections. In particular, the aim is to understand how different parameters like shape (α), chirality (C), and noise (D_r) influence particle motion in both steady and time-dependent flow fields.

This section presents the key results observed from simulations. The first focus is on the effect of particle shape.

4.1 Aggregation of Ellipsoidal Particles

When $\alpha = 0$, the particles are spherical and do not show aggregation. The distribution remains uniform in space because of volume-preserving dynamics. Figure 3 shows the distribution for $\alpha = 0$, where the particles remain spread uniformly.

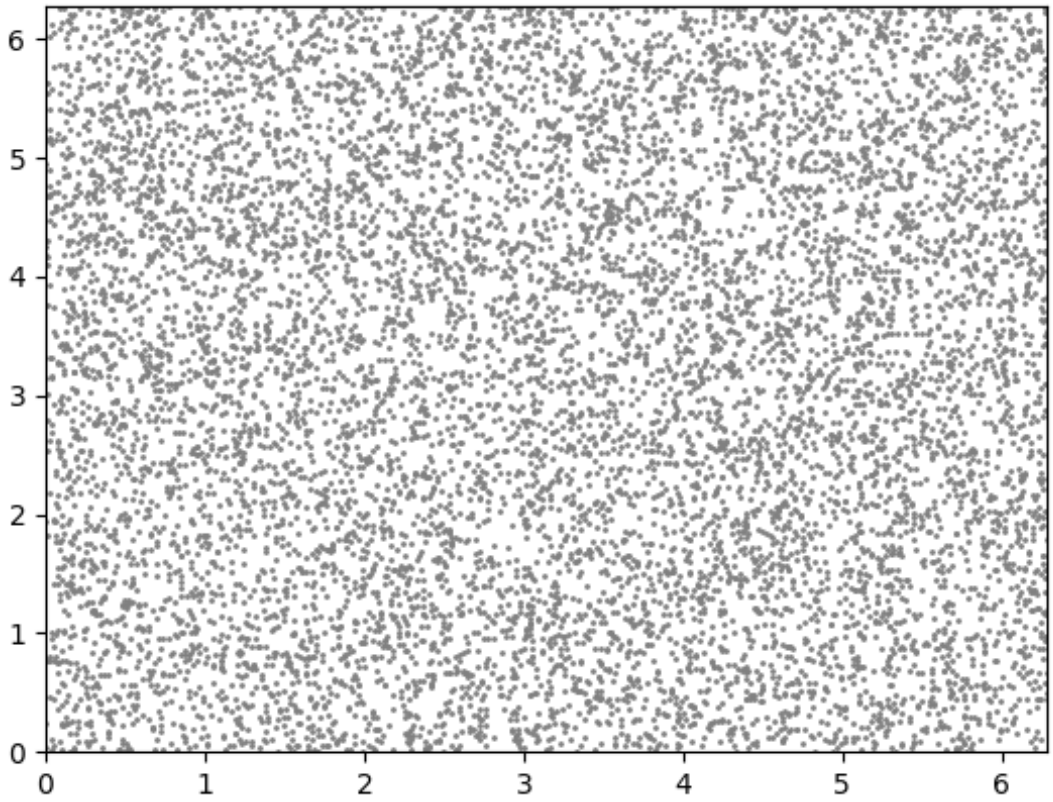
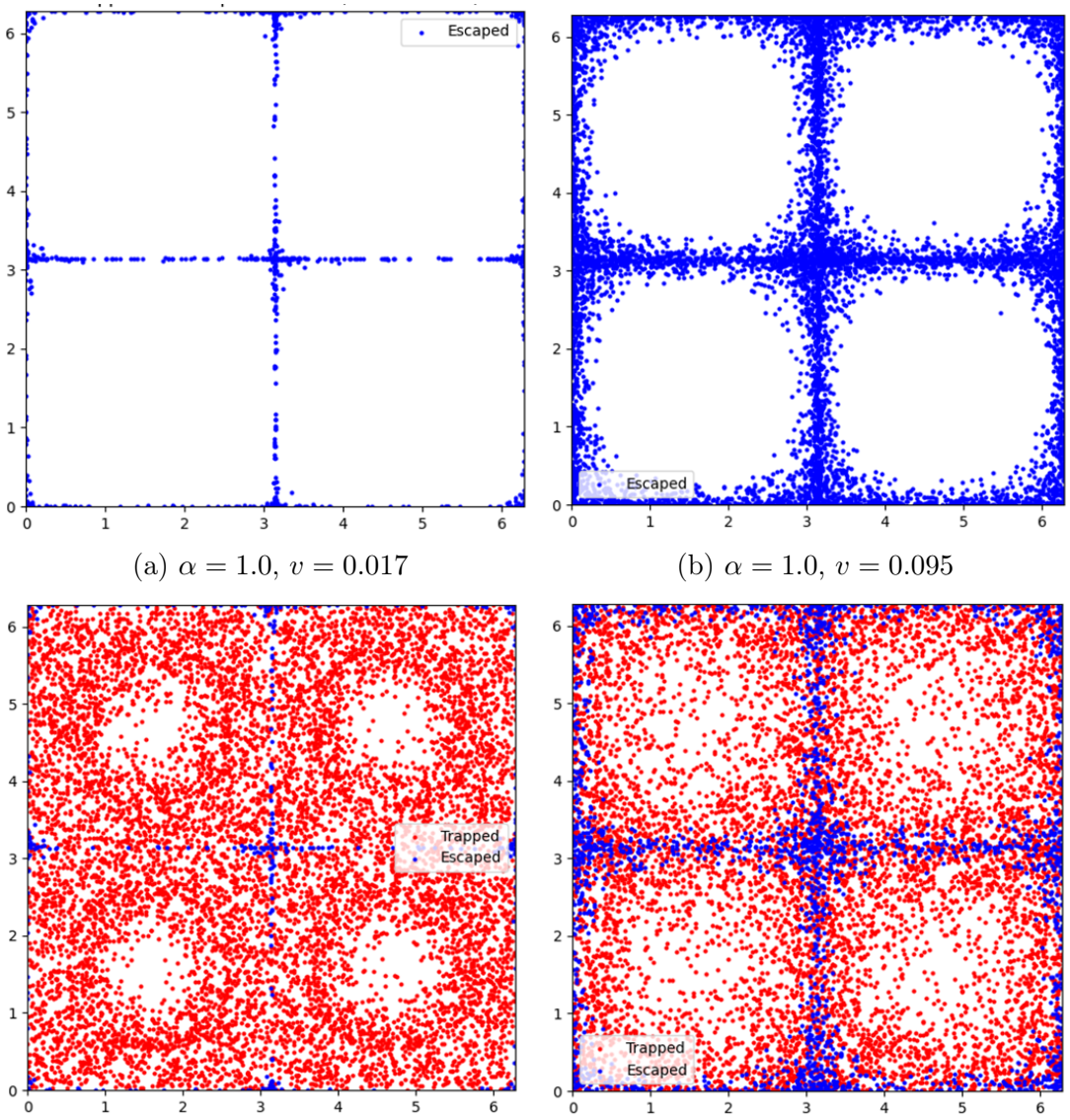


Figure 3: Particle distribution for $\alpha = 0$ (spherical particles). No aggregation is observed.



(a) $\alpha = 1.0$, $v = 0.017$

(b) $\alpha = 1.0$, $v = 0.095$

(c) $\alpha = 0.9$, $v = 0.016$

(d) $\alpha = 0.95$, $v = 0.095$

Figure 4: Particle aggregation is observed only when α approaches 1 and v approaches 0. Particles accumulate near separatrices. Blue indicates particles which move between cells; Red are trapped particles.

For nonzero α, especially when α is close to 1 and v is small, particles start to accumulate near the separatrices of the flow. Figure 4 shows the particle distributions for different combinations of α and v.

4.2 Dependence of Minimum Swimming Velocity on Particle Shape

One of the objectives in this study is to find the minimum swimming velocity needed for a particle to escape from a vortex cell. This depends on the shape of the particle, which is described by the eccentricity parameter α.

For each value of α, the minimum velocity was calculated by checking the smallest swimming speed at which at least 80% of the particles could leave their starting vortex within. This gives a clear idea of how shape affects the ability of the particle to overcome the trapping effect of the flow.

The results are shown in Figure 5. The plot shows that when the particle is more elongated (higher α), the minimum velocity needed for escape is lower. This happens because elongated particles can align better with the flow and use it to move across the cell. On the other hand, when the particle is close to spherical (α is small), it needs more swimming speed to escape the vortex.

The plot shows a trend where shape and swimming speed together decide how easy it is for particles to escape.

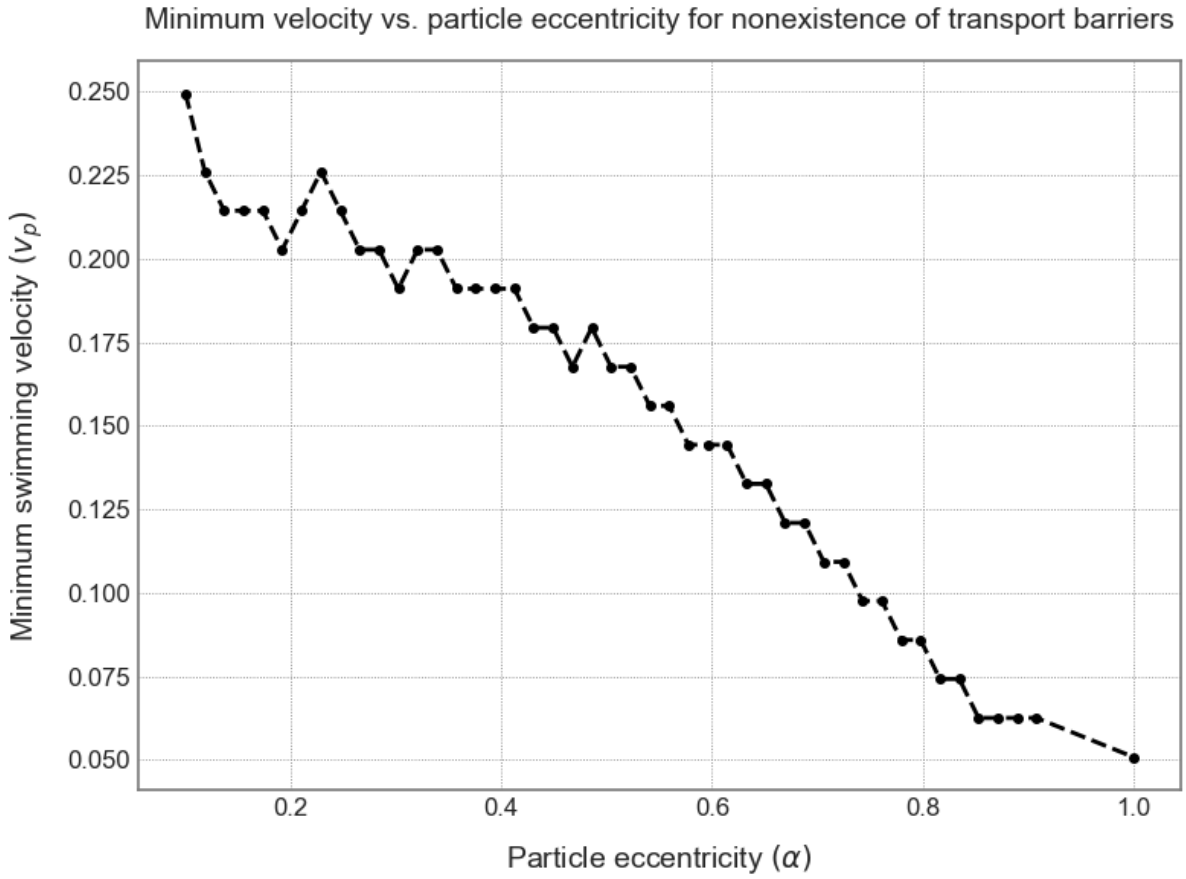


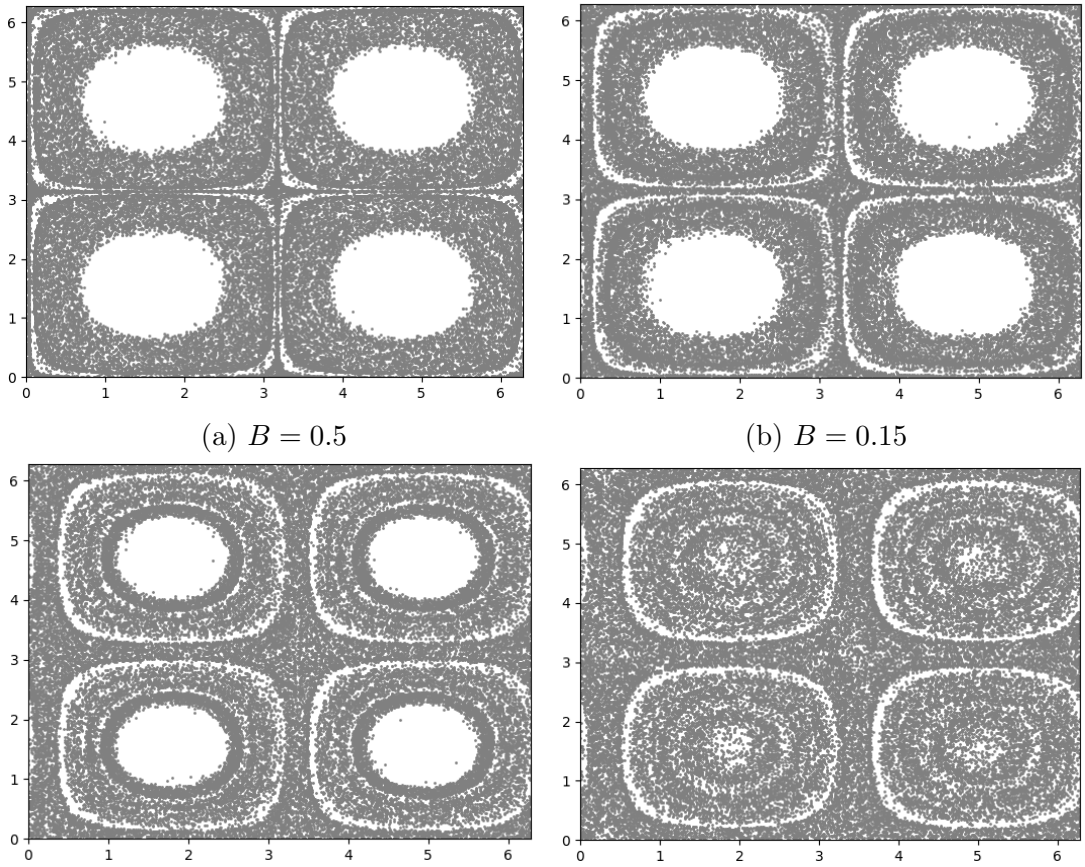
Figure 5: Minimum swimming velocity needed for particles to escape the vortex cell, plotted as a function of the particle eccentricity (α). Elongated particles need less swimming speed to escape, while spherical particles need more.

4.3 Effect of Time-Dependent field

When time-dependence is introduced into the flow (via a nonzero B), the position of the separatrices begins to oscillate. This weakens their ability to confine particles and allows them to cross into neighboring vortex cells. The regions near separatrices become unstable, where transport between cells is enhanced.

As the amplitude B is increased, the width of this chaotic region grows, providing more pathways for particle escape even at low self-propulsion speeds.

Figure 6 shows how the spatial distribution of particles changes for increasing values of B. The figures demonstrate that the particles gradually fill a larger portion of the domain as B increases, highlighting the broadening of the chaotic region.



(a) $B = 0.5$ (b) $B = 0.15$

(c) $B = 3.5$ (d) $B = 5.0$

Figure 6: Particle distributions for increasing values of B. As the flow oscillation amplitude grows, the chaotic region widens, allowing particles to explore a larger portion of the domain.

4.4 Effect of Rotational Diffusion

Self-Propelled particles immersed in water constantly collide with fluid molecules.These random interactions cause the particle to change its orientation over time. This process is modeled as **rotational diffusion**, where the orientation angle θ is affected by a stochastic term.

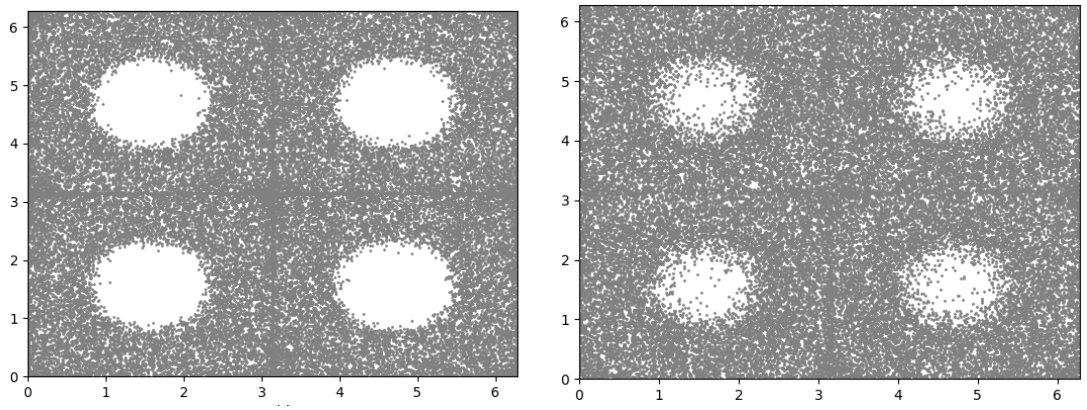
$$\langle \eta(t)\eta(t') \rangle = 2D_r\delta(t - t') \tag{10}$$

This is included in simulations via a Gaussian white noise term added to the angle θ:

$$\dot{\theta} = -2\cos x \cos y \sin(2\theta) + \sin x \sin y + \sqrt{2D_r}\,\eta(t) \tag{11}$$

It plays a key role in preventing long-term alignment with flow structures, and helps particles escape traps by randomly reorienting them. In the absence of diffusion, elongated particles tend to align with the flow and accumulate near separatrices. However, when rotational noise is present, this alignment is disrupted, and particle clustering is weakened.

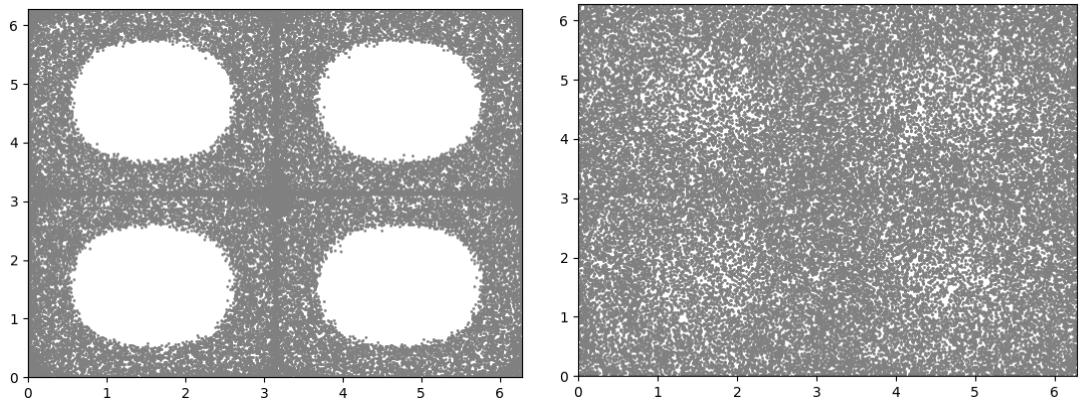
Figure 7 shows the long-time distribution of particles in the steady flow case ($B = 0$) for different values of the rotational diffusion coefficient D_r. As D_r increases, the distribution becomes more uniform and aggregation is suppressed.



(a) $D_r = 0$ (b) $D_r = 0.01$

Figure 7: Effect of rotational diffusion on particle distribution for $\alpha = 1.0$, $v = 0.015$, and $B = 0$. Noise reduces aggregation. Snapshots taken at t = 100.

To further investigate the role of flow time-dependence, simulations were performed with $B \neq 0$. In this case, the separatrices oscillate, and particles are able to cross between vortex cells more easily. Figure 8 compares the particle distributions for the different D_r values in a time-dependent flow.



(a) $D_r = 0$, $B = 0.1$ (b) $D_r = 0.01$, $B = 0.1$

Figure 8: Rotational diffusion in time-dependent flow. Mixing is enhanced due to oscillating separatrices.

Our investigations indicate that for $D_r \lesssim U_0/L$, [3] the elliptic regions begin to empty out due to the effect of rotational diffusion. As the noise intensity increases further, this effect becomes less pronounced, and eventually, the particle distribution becomes nearly uniform across the domain.

4.5 Effect of Chirality on Particle Dynamics

4.5.1 Biological and Physical Origins of Chirality

Chirality refers to a built-in tendency of a particle to rotate in a particular direction as it moves forward. This behavior is observed in many biological systems. For instance, sperm cells and certain bacteria like *E. coli* follow curved or circular paths due to internal asymmetries in their propulsion or body structure.

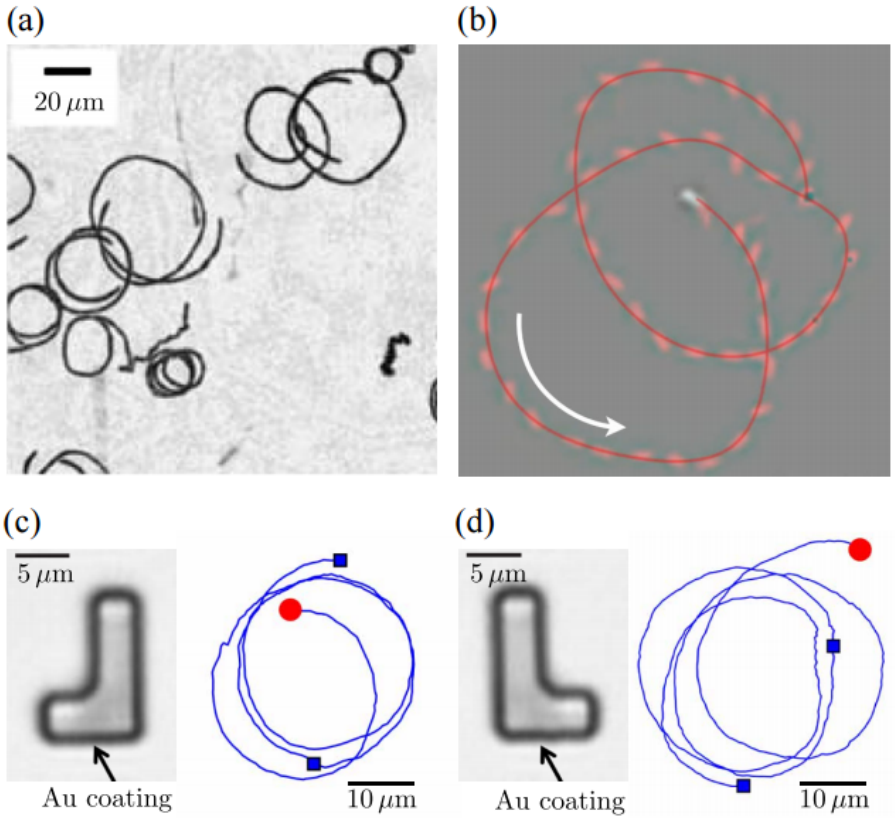


Figure 9: Examples of chiral trajectories in biological and artificial microswimmers. (a, b) Circular swimming patterns of *E. coli* bacteria . (c, d) Self-propelled artificial microswimmers driven by self-diffusiophoresis, exhibiting either dextrorotatory (clockwise) or levorotatory(anti-clockwise) motion depending on particle asymmetry. Taken from [2].

Biological sources of chirality may include:

- Uneven beating of flagella
- Asymmetric propulsion mechanisms
- Interactions with nearby surfaces or boundaries

In the model, chirality is represented by a constant angular velocity term C in the orientation dynamics of the particle.

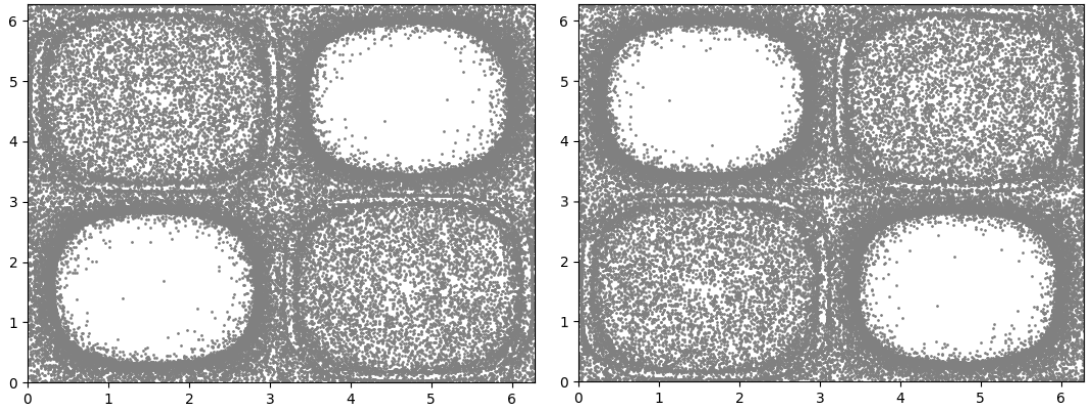
$$\dot{\theta} = -\alpha \cos x \cos y \sin(2\theta) + \sin x \sin y + C \tag{12}$$

A positive C causes anti-clockwise rotation, while a negative C causes clockwise rotation. This additional term influences how the particle interacts with the flow field.

4.5.2 Chirality in Vortex Flows

In a steady vortex flow, chiral particles tend to accumulate in regions where the flow rotation matches their turning direction. Non-chiral particles usually show symmetric distributions around vortex centers. Chirality breaks this symmetry.

Figure 10 shows how positive and negative chirality influence the spatial distribution of particles in the flow. Clustering is no longer symmetric and is biased by the sign of C.



(a) $C > 0$ (counterclockwise) (b) $C < 0$ (clockwise)

Figure 10: Particle distributions in steady vortex flow for non-zero chirality values. Chirality breaks left-right symmetry.

4.5.3 Chiral Particle Trajectories in Phase Space

The effect of chirality can also be seen in the particle's motion in phase space. Here, phase space refers to the three-dimensional space formed by the spatial coordinates (x, y) and the orientation angle θ.

For non-zero C, the orientation of the particle winds continuously over time, resulting in helical or spiral paths in phase space. These patterns are distinct for clockwise and counterclockwise turning particles.

Figure 11 shows typical 3D trajectories for different chirality values.

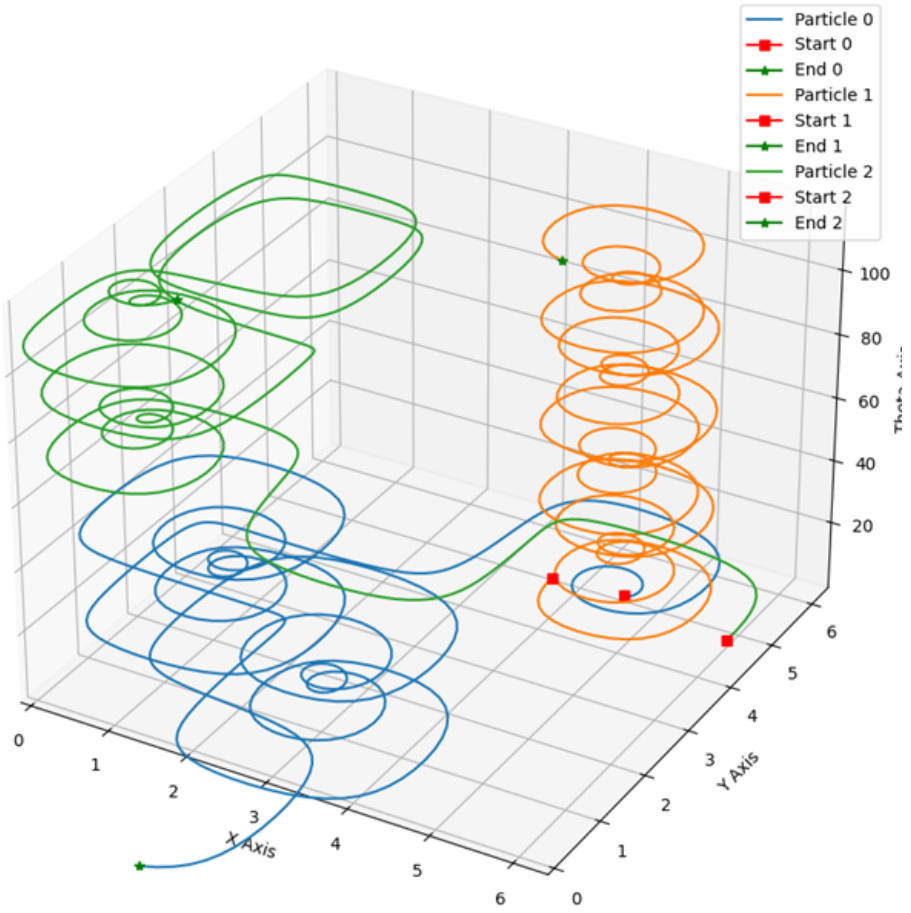


Figure 11: Trajectories of chiral particles in 3D phase space (x, y, θ) .

4.6 Vorticity

Vorticity is a quantity that represents local rotation of the fluid. It is defined as:

$$\omega = \nabla \times \vec{v} = \frac{\partial v_y}{\partial x} - \frac{\partial v_x}{\partial y}$$

In two-dimensional flows, vorticity is a scalar. In the cellular flow considered here, the vorticity field forms a regular checkerboard-like pattern, with alternating regions of positive and negative rotation.

Figure 12 shows the vorticity field across the domain.

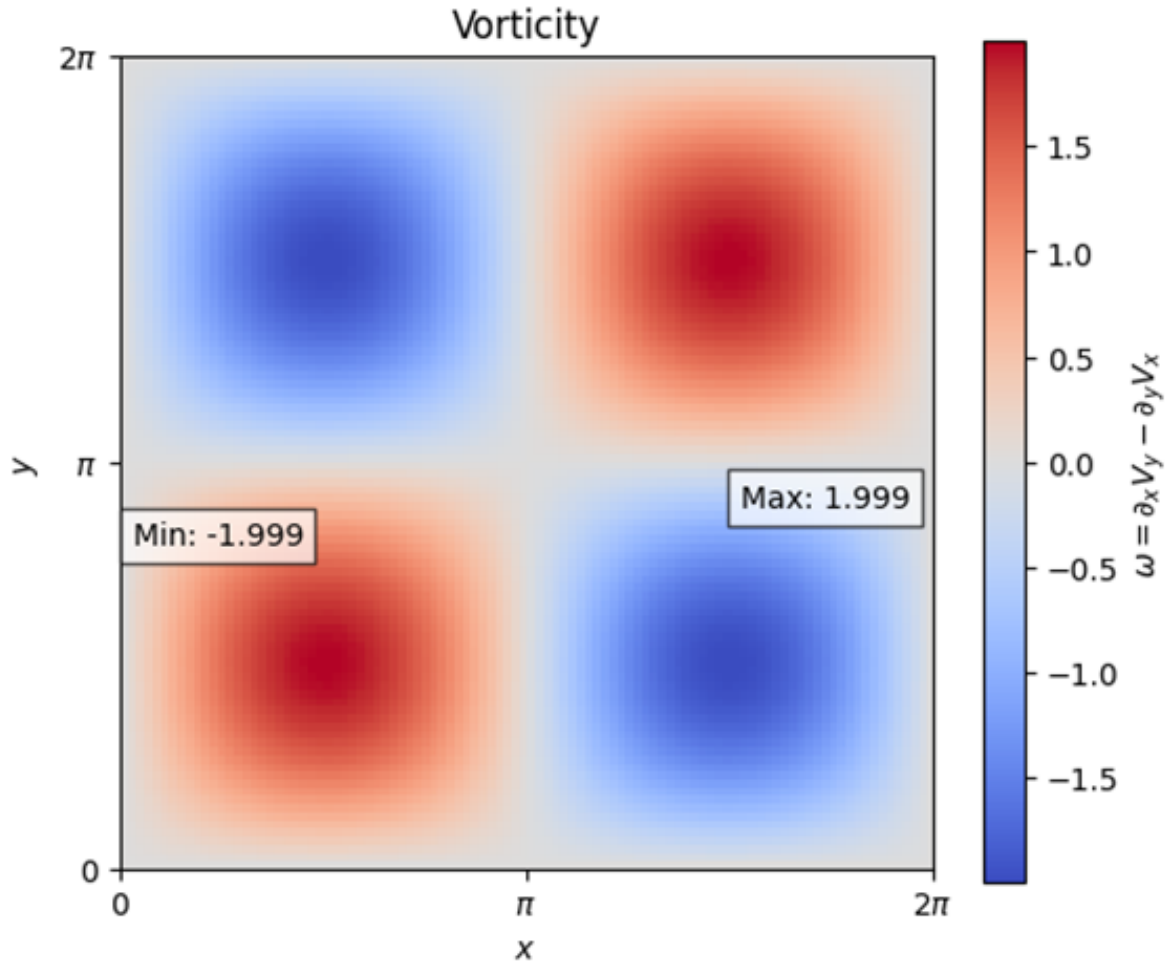


Figure 12: Vorticity field for the cellular flow over domain $[0, L] \times [0, L]$. Red and blue regions correspond to opposite signs of vorticity.

4.6.1 Vorticity Distribution

One way to understand the effect of chirality on particle dynamics is by looking at the vorticity values encountered along their trajectories. Since the flow is structured, particles do not sample the flow uniformly — especially when chirality is present. To capture this bias, the vorticity ω is recorded at the position of each particle during its motion, and the probability distribution $P(\omega)$ is constructed.

This distribution gives a direct picture of where the particles tend to spend their time: regions with higher vorticity probability indicate flow structures where particles are more likely to be trapped or guided.

However, even without chirality, particles might sample the flow unevenly due to geometric factors or flow structure. For example, if particles are uniformly spread out, some flow regions (like vortex edges) may naturally contain more particles simply because they occupy more space. To remove this effect, the distribution is normalized by the vorticity distribution obtained from particles uniformly distributed over the domain, denoted $P_0(\omega)$.

The ratio

$$R(\omega) = \frac{P(\omega)}{P_0(\omega)}$$

highlights the true bias introduced by chirality.

Figure 13 shows this ratio for two cases: a non-chiral system and a weakly chiral system. Both simulations were performed with the same self-propulsion speed $v_p = 0.15$ and alignment parameter $\alpha = 0.99$.

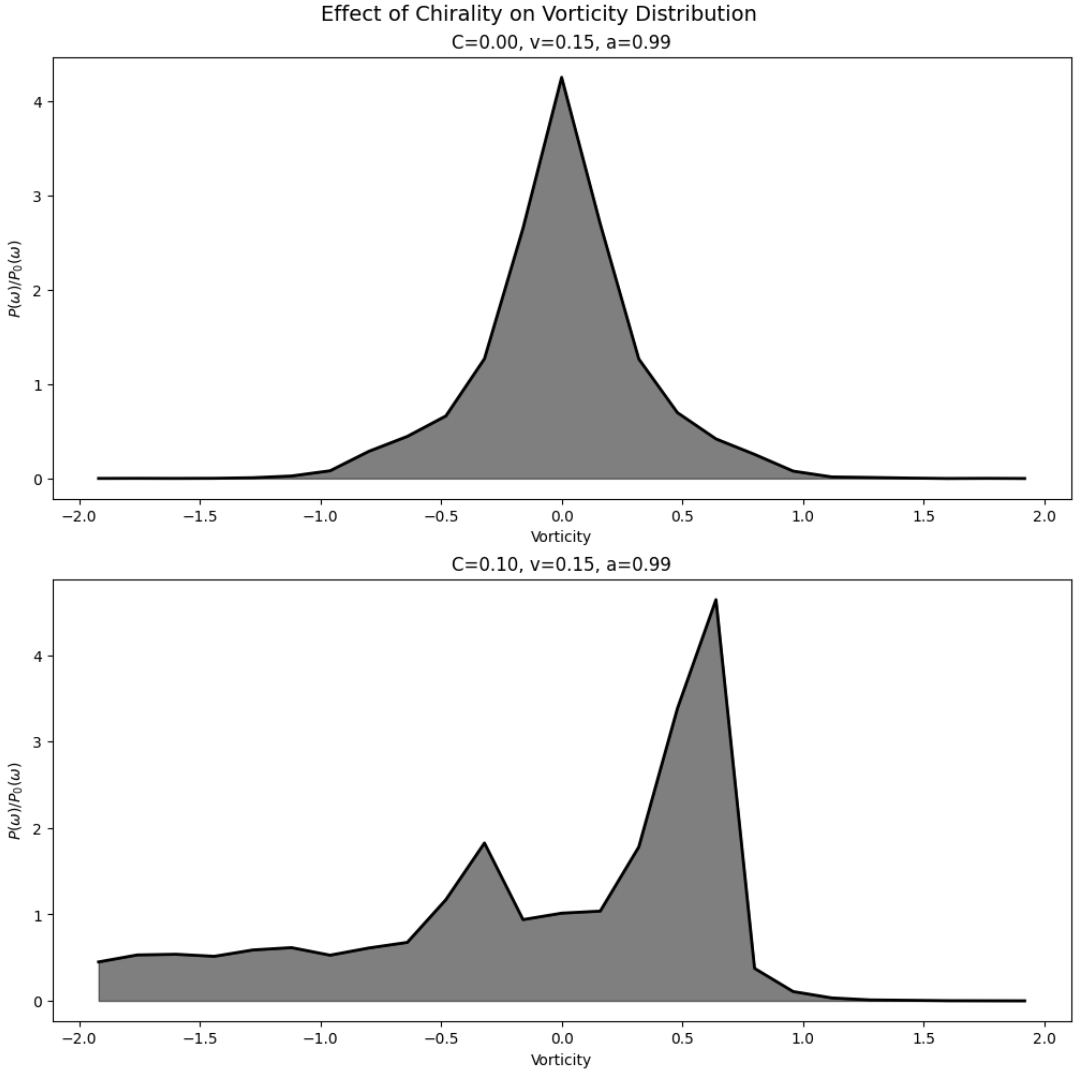
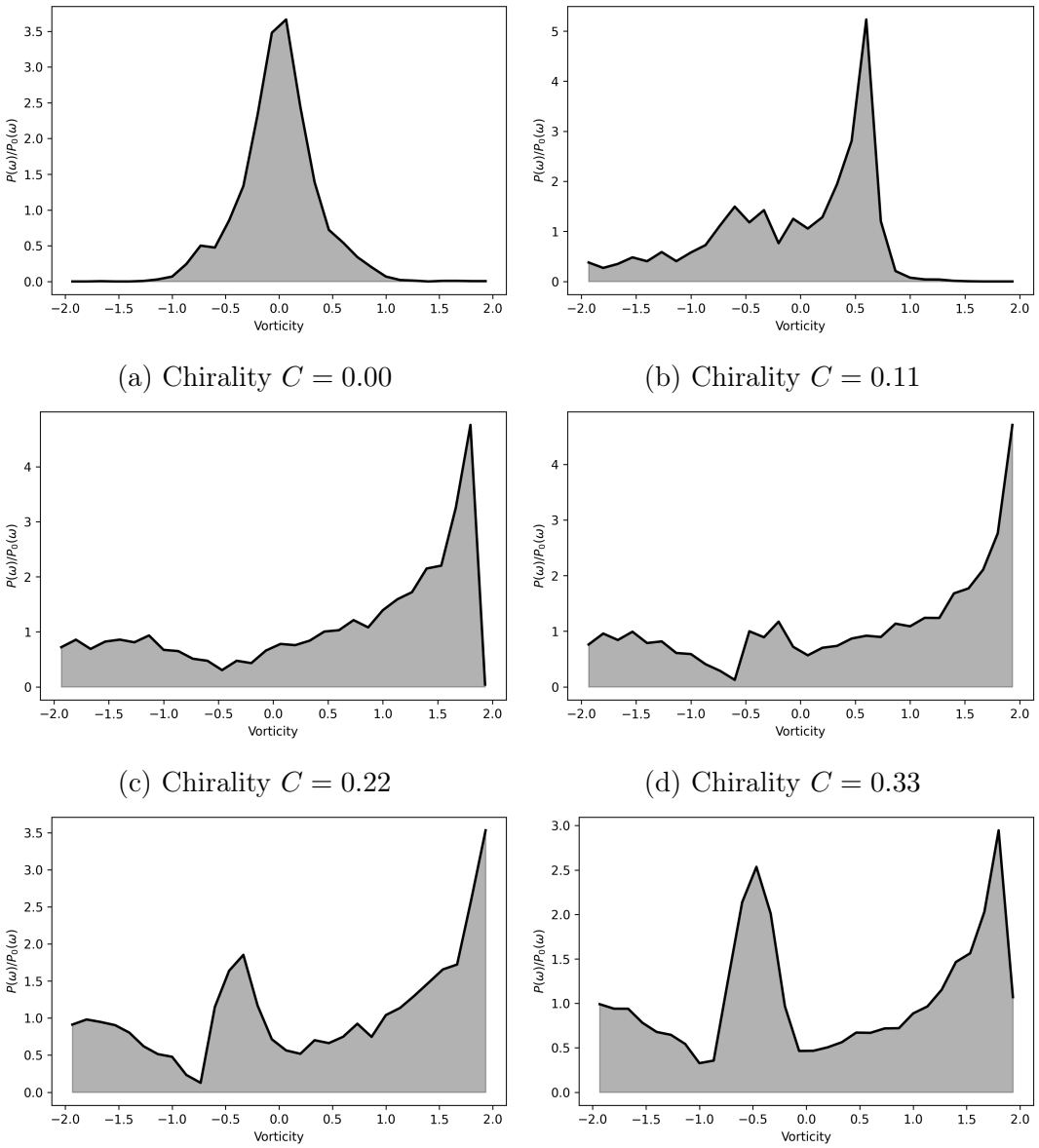


Figure 13: Effect of chirality on the vorticity distribution sampled by particles. The top panel shows the ratio $P(\omega)/P_0(\omega)$ for a non-chiral system ($C = 0.00$), where the distribution remains symmetric and peaked around zero, reflecting uniform and unbiased sampling. The bottom panel shows the same ratio for a weakly chiral system ($C = 0.10$).

So far, the vorticity distribution was shown for just two cases: one where the chirality is zero ($C = 0$) and one where the particle has finite chirality. To better understand the full effect of chirality, the distribution was computed for a range of values from $C = 0$ to $C = 1$.

Figure 14 and Figure 15 shows how the vorticity distribution changes as chirality increases. When $C = 0$, the particles mostly sample regions near zero vorticity, which is expected since they follow the flow structure. As C increases, the distribution shifts, and particles are seen more often in regions of higher vorticity. This shift becomes more clear as the chirality gets closer to 1.

This shows that the presence of chirality leads to a strong preference for certain regions in the flow, and the effect is gradual and smooth as the chirality is increased.



(a) Chirality $C = 0.00$

(b) Chirality $C = 0.11$

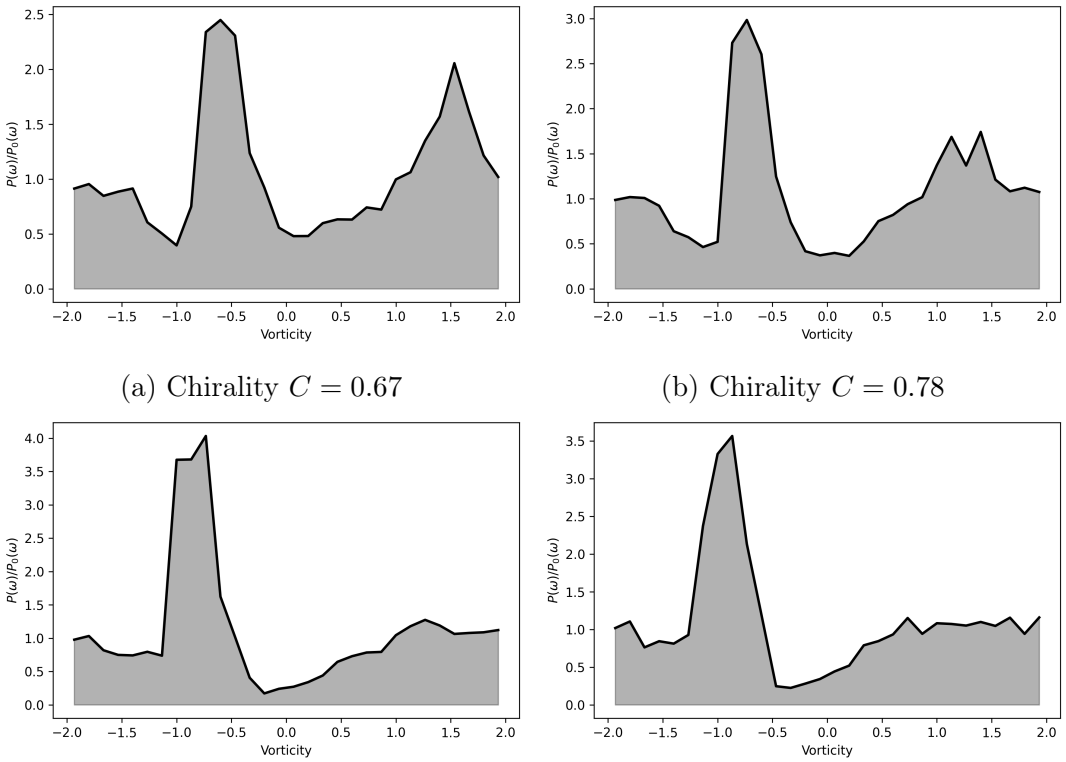
(c) Chirality $C = 0.22$

(d) Chirality $C = 0.33$

(e) Chirality $C = 0.44$

(f) Chirality $C = 0.56$

Figure 14: Vorticity distributions for chirality values $C = 0.00$ to $C = 0.56$.



(a) Chirality $C = 0.67$

(b) Chirality $C = 0.78$

(c) Chirality $C = 0.89$

(d) Chirality $C = 1.00$

Figure 15: Vorticity distributions for chirality values $C = 0.67$ to $C = 1.00$.

4.6.2 Effect of Chirality on Vorticity Distribution

When the chirality is zero ($C = 0$), the particles do not have any preferred turning direction. Because of this, they explore the flow symmetrically, spending equal amounts of time in both clockwise and counter-clockwise vortices. The vorticity distribution is symmetric in this case.

When chirality is added, the behavior changes. The turning direction of the particle starts to either match or oppose the rotation of the flow, depending on which vortex it is in. In vortices where the chirality and flow rotation match, the particle stays longer. In the opposite vortices, it leaves sooner. This creates an imbalance in how the particles sample the flow.

This difference shows up clearly in the vorticity distribution, which shifts away from being symmetric and becomes more biased as the chirality increases.

5 Conclusion

In this project, the motion of ellipsoidal self-propelled particles in a two-dimensional flow was studied. The focus was on understanding how the shape of the particle, chirality, and flow conditions affect their transport and distribution.

It was seen that for spherical particles, the distribution remains uniform, but once the shape is changed, particles tend to gather in certain regions of the flow. Chirality adds another layer to this behavior by pushing the particles to stay longer in vortices that match their own turning direction. The study also looked at the effect of random rotation

(rotational diffusion) and how time-dependent flows allow particles to move across flow barriers.

Numerical simulations were used to explore different cases, and the results give a clear picture of how shape and chirality work together to change the way particles behave in a flow. The vorticity distribution and escape behavior were used to understand these effects in more detail.

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